How College Students Actually Spend Their Meal Plan Dollars

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Research Questions

1. **What proportion of meal plan dollars actually goes toward dining halls vs. fast food or retail outlets?**
2. **Do meal plan spending patterns change by student demographics (e.g., year in school, income status)?**
3. **How does national food-away-from-home spending compare with student-reported behavior?**

Motivation

Meal plans and college food spending represent a large financial commitment. Students often feel they either overspend early or struggle later in the term. By comparing national food spending data with student survey responses, this project explores whether real spending patterns match student expectations and perceived value.

Dataset

USDA Food Expenditure Series (ERS)
Source: https://www.ers.usda.gov/data-products/food-expenditure-series

Due to lack of public access to the originally proposed UNC transaction dataset, I transitioned to the USDA Food Expenditure Series, which provides structured CSV files with national food spending by outlet type.

This dataset tracks national food spending over time, including food-away-from-home purchases. Each row represents a monthly or yearly spending record. Columns include Year, spending totals, and category breakdowns.

The dataset required cleaning because column names were long and values included commas that prevented direct numeric analysis.

Student Food Spending Survey (Kaggle)
Source: https://www.kaggle.com/datasets/mlomuscio/student-food-survey

This dataset contains student responses about dining behavior, weekly spending, grade level, and perception of affordability. Each row represents one student response and helps explain behavior behind the spending patterns.

Average College Student Food Spending
Source: https://educationdata.org/average-monthly-food-spend-college-student

This dataset provides national benchmark statistics on student food spending. It is used as contextual comparison rather than transaction-level analysis.

Method

I loaded both datasets using pandas and performed cleaning steps programmatically. For the USDA dataset, I identified relevant columns, removed commas from numbers, converted values to numeric format, and created a Date column from the year and month fields.

For the survey dataset, I standardized column names and created a consistent 'Year in School' field from the original Grade column. After cleaning, I generated summary statistics, checked missing values, and created multiple visualizations to explore patterns. These visualizations include Figures 1–4, which connect cleaned variables directly to the research questions.

EDA RESULTS

Dataset Size and Structure

Dataset 1 — USDA Food Expenditure Data

This dataset contains national food spending records collected by the USDA.

Rows: **28**, Columns: **27**

Each row represents one year of food expenditure data.
Columns include multiple categories of food spending, such as:

• Total food-at-home expenditures
• Food sales at full-service restaurants
• Food sales at limited-service restaurants
• Food sales at drinking places
• Food sales at retail stores

During preprocessing, my code extracts and standardizes: **Food Away From Home** - spending on prepared food and dining locations, **Food At Home** - grocery-style spending, **Year**, and a generated **Date**

column. These cleaned variables are used for all analysis and visualizations. Figures 3 and 4 use these cleaned USDA variables to show long-term national trends in food-away-from-home spending.

Dataset 2 — Student Food Spending Survey

This dataset contains survey responses from college students about their dining habits and food spending behavior. It helps explain the "why" behind the trends seen in the USDA spending patterns.

Rows: 167, **Columns:** 14

Each row represents one student response.

Important variables used in my analysis:

• Grade → renamed to **Year in School** in code
• Money - weekly spending amount
• DHBreakfast, NDHMeals, DHBoxes - dining behavior indicators
• Nutrition - perception-based measure

This dataset provides behavioral context behind the spending trends seen in the USDA data.

Missing Data Analysis

I checked missing values using:

```
df.isnull().sum()
df.isnull().mean() * 100
```

This allowed me to examine both total missing values and percentage missingness for each dataset.

Dataset 1 — USDA Food Expenditure Data

• All major spending columns contain **0% missing values**
• The Year column is fully complete
• No financial variables required row removal

Because this dataset comes from a structured government source, it was already highly complete. The main preprocessing step involved converting numeric strings with commas into usable numeric values rather than handling missing data.

Handling Approach: No rows were removed. Spending values were converted to numeric format. Date column was generated from the Year field. Since there was no significant missingness, the dataset could be used directly after cleaning.

Dataset 2 — Student Food Spending Survey

• Grade → 1 missing value (~0.6%)
• Athlete → 1 missing value (~0.6%)
• Activities → 1 missing value (~0.6%)
• DHBoxes → 2 missing values (~1.2%)
• NDHBoxes → 3 missing values (~1.8%)
• Nutrition → 1 missing value (~0.6%)
• Money → 26 missing values (~15.6%)

The Money column has the largest amount of missing data, which is expected because survey respondents sometimes skip financial questions.

Handling Approach:

• Behavioral categorical variables were kept as-is since missingness was minimal
• The Money column was converted to numeric values using `pd.to_numeric(errors="coerce")`
• Rows with missing Money values were excluded only from boxplot visualizations, not from the entire dataset

This approach preserves as much survey information as possible while ensuring accurate statistical summaries.

Variables of Interest

My analysis focuses on variables that directly connect spending trends with student behavior.

From USDA Dataset (after cleaning)

• *Food Away From Home:* Represents restaurant-style spending. Acts as a proxy for dining hall or campus food usage.

• *Food At Home*: Represents grocery-style spending and provides a comparison baseline.

• *Year / Date:* Used to examine long-term spending trends over time.

These fields are created programmatically in `clean_usda()` inside `mealplan_analysis.py`.

From Survey Dataset

• *Year in School (derived from Grade):* Used to compare behavior across student groups.

• *Money:* Numeric weekly spending variable used in boxplots.

• *Dining Behavior Columns (DHBreakfast, NDHMeals, DHBoxes):* Provide insight into how often students rely on dining services.

These variables align exactly with the columns referenced in the code's visualization functions.

The survey spending variable (Money) is visualized in Figure 1, which shows how weekly spending varies across grade levels. The distribution of survey responses by class year is also shown in Figure 2 to provide context for how representative each group is within the dataset.

Summary Statistics

Quantitative Variables — USDA Spending

Summary statistics are generated using:

```
df.describe()
```

Food Away From Home spending shows gradual increases across years, reflecting rising reliance on prepared food purchases. Food At Home spending remains comparatively stable, providing a useful baseline for comparison.

Quantitative Variables — Survey Spending (Money)

• Mean and median values show moderate weekly spending.
• Wide range indicates strong variation in student behavior.

This variation becomes clearer in Figure 1, where several higher-value outliers appear among upper grade levels. This variability supports the idea that meal plan usage differs significantly between individuals.

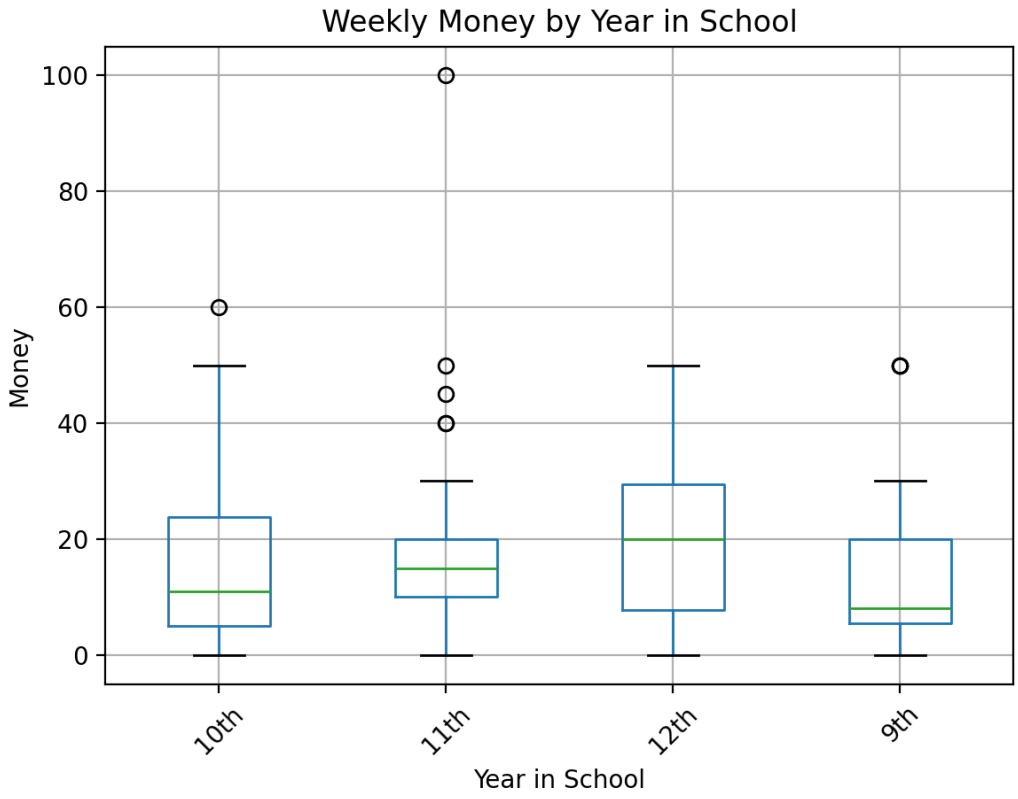
Categorical Variable — Year in School

Derived from the original `Grade` column.

Unique values include: 9th, 10th, 11th, 12th

Figure 2 visualizes how frequently each grade level appears in the dataset, which helps interpret later comparisons. These categories are used directly in the bar chart visualization produced by the analysis script.

Figure 1

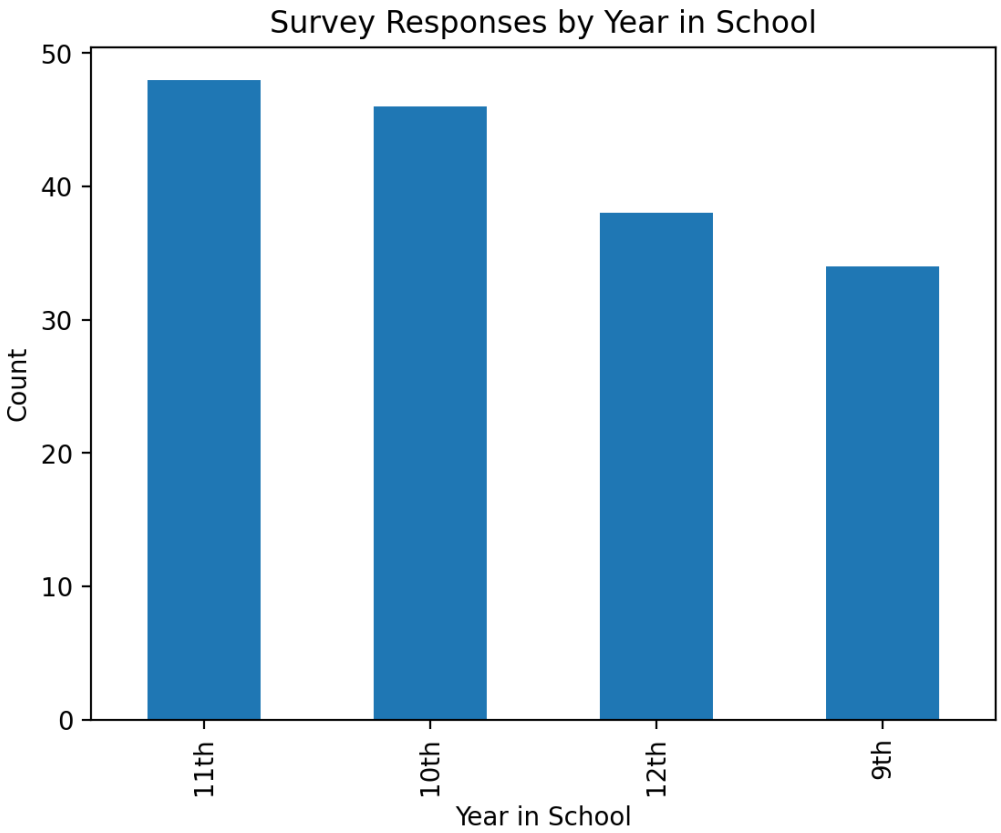


X-axis: Year in School
Y-axis: Money ($)

Caption: Spending distributions vary across grade levels, with upperclassmen showing slightly higher variability and some larger outliers.

Explanation:
This boxplot shows the distribution of weekly spending (Money) grouped by Grade from the student food survey dataset. I chose a boxplot because it clearly displays medians, spread, and outliers across groups, which helps compare behavior between class levels. The plot was produced using matplotlib after grouping survey responses by year in school. The main takeaway is that spending is not uniform across grades. Some upper-year students report much higher spending, suggesting differences in independence or dining habits as students progress through school.

Figure 2

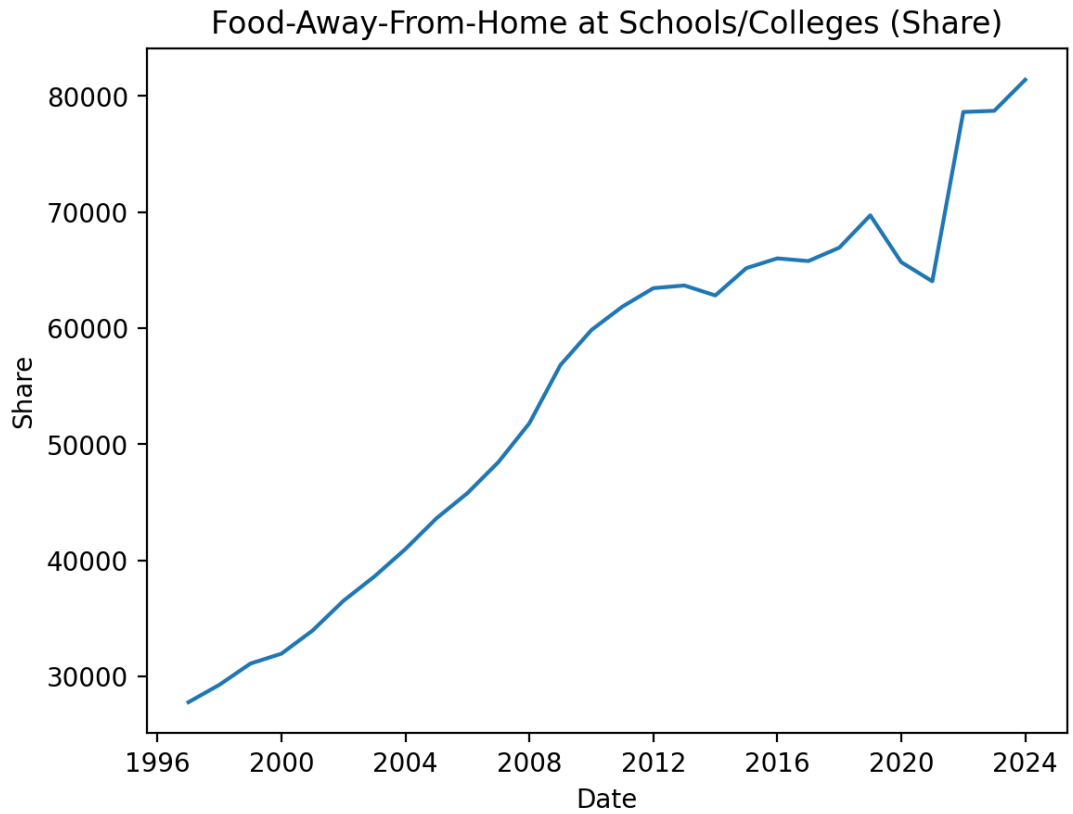


X-axis: Year in School
Y-axis: Count of Responses

Caption:
Most survey responses come from 10th and 11th grade students, creating slight class-level imbalance in the dataset.

Explanation:
This bar chart shows how many survey responses came from each grade level. I chose a bar chart because the variable is categorical and I wanted to visualize representation across groups. The visualization was generated using value counts on the Grade column. This plot is important because it provides context for later comparisons. Since some grade levels have more responses than others, differences in behavior should be interpreted with that imbalance in mind.

Figure 3

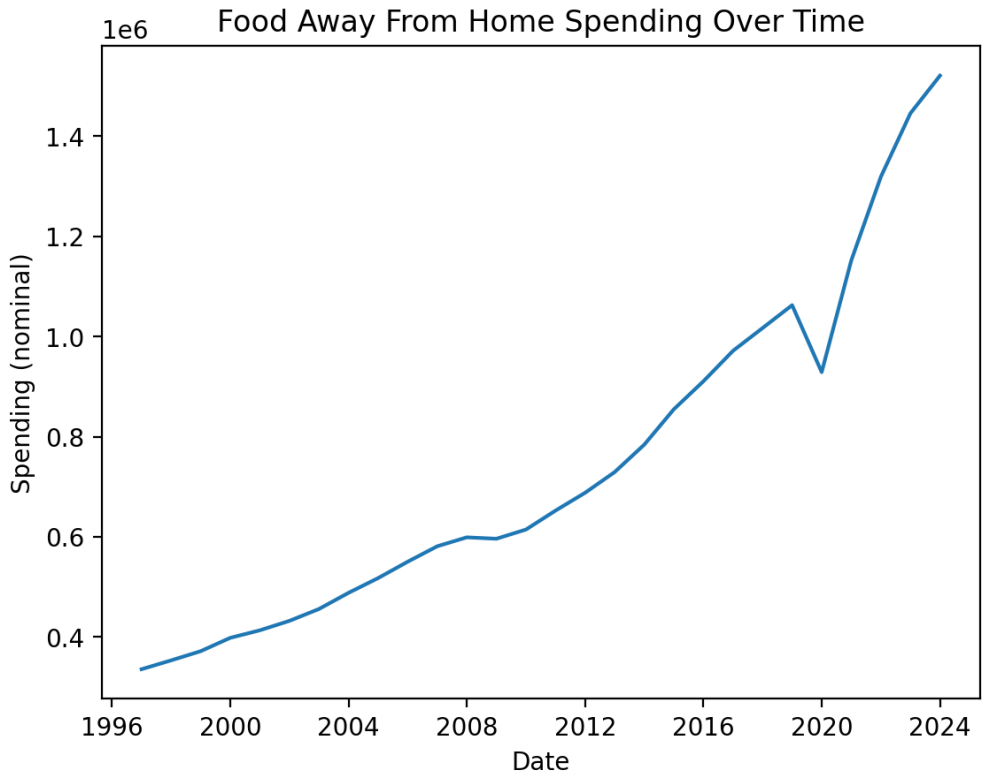


X-axis: Date
Y-axis: Share of Spending

Caption:
National spending share on food away from home at schools and colleges shows an overall upward trend over time.

Explanation:
This line plot uses the USDA dataset to show how spending share on prepared food at schools and colleges has changed over time. I selected a line chart because the dataset represents a time series and trends are easier to observe when plotted sequentially. The visualization was created after converting messy numeric strings into numeric values and building a Date column from the year variable. The main takeaway is that spending on prepared food environments has steadily increased, which provides important national context for understanding dining plan usage patterns among students.

Figure 4



X-axis: Date
Y-axis: Spending (nominal)

Caption:
Total spending on food away from home has risen consistently, indicating growing reliance on prepared food purchases.

Explanation:
This line chart displays nominal spending on Food Away From Home from the USDA dataset. I chose this visualization because it highlights long-term financial trends and helps compare national behavior with survey-based student patterns. The chart was generated after cleaning messy column names and converting values that originally contained commas into numeric format. The upward trend suggests that prepared food spending has become more common nationally, which supports the idea that dining plans and campus food systems reflect broader economic trends rather than isolated student behavior.

Multiple Dataset Relationship Summary

The two datasets support different parts of the same question.

The USDA dataset shows real national spending trends over time, especially food spending away from home versus food at home. This gives a baseline for how people typically use money on prepared food.

The student food survey dataset focuses on individual behavior and perception. It includes variables like Grade (converted to Year in School), weekly spending (Money), and dining behavior indicators such as DHBreakfast, NDHMeals, and DHBoxes.

Instead of merging individual rows, I compare patterns across both datasets. For example:

• Food Away From Home trends in USDA data compared with student dining behavior indicators
• National spending trends compared with student-reported weekly spending
• Year in school compared with survey behavior patterns

Using both datasets together strengthens the analysis because one shows large-scale financial trends while the other explains student decision-making. Figures 3 and 4 provide the national spending context, while Figures 1 and 2 illustrate student-level behavior and representation within the survey dataset.

Challenge Goals Update

My project fulfills both the **Multiple Datasets** and **Messy Data** challenge goals.

Originally, I planned to combine a university transaction dataset with survey responses. However, the UNC dining tables were embedded inside a research article and were not available in CSV format, which made automated analysis unrealistic.

Because of this, I switched to the USDA Food Expenditure dataset. While this dataset is structured, it still required significant preprocessing that satisfies the messy data challenge:

• Extremely long and inconsistent column names
• Numeric values stored as strings with commas
• Creation of a new Date column from year values
• Programmatic renaming of variables to usable formats

Instead of joining rows directly, I compare datasets at the trend level. The USDA dataset provides financial context, while the survey dataset explains how students interact with dining systems.

This adjustment keeps the project realistic while still meeting both challenge goal requirements.

Work Plan Evaluation

My original estimates were mostly accurate, but data sourcing took longer than expected.

Finding a usable structured dataset required extra time because my original transaction source was not machine-readable. Switching to the USDA dataset added about two extra hours of work early in the project.

Data cleaning was actually more complex than expected because the USDA file required numeric conversion, column matching, and creation of new variables through code. EDA tasks like generating summaries and visualizations were close to my original estimate at about four hours.

Remaining tasks include:

• Final visualization formatting
• Proofreading and formatting

I estimate around another hour to finalize everything.

TESTING

I tested my EDA code in several ways to make sure results were accurate.

1. Assert Statements

I used assert statements to verify cleaning steps worked correctly. For example:

```
assert cleaned["Food Away From Home"].isnull().sum() == 0
```

This ensured numeric conversion and missing data handling behaved as expected after cleaning messy numeric strings.

2. Small Test DataFrames

Before running the full datasets, I created small test DataFrames with known values. This helped confirm that: Comma removal worked correctly, numeric conversions succeeded, and that Date column creation was valid.

3. Summary Statistic Checks

I used `df.describe()` to verify that spending values stayed within realistic ranges. This helped catch formatting issues caused by messy numeric strings before visualization.

COLLABORATION

I completed this project independently.

Resources consulted:

- USDA dataset documentation
- Kaggle dataset page for the student survey
- Pandas documentation
- Matplotlib documentation
- CSE 163 lecture notes and examples

No collaboration with other students occurred during this stage of the project.